Assignment 5

Name: ASHMIT KUMAR

Course: BTECH ECE

QUANTUM UNIVERSITY

Task 1: Agile Development Models

1. **Product Identified:** "EcoTrack" – A mobile application that calculates and gamifies a user's daily carbon footprint based on their travel and shopping habits.

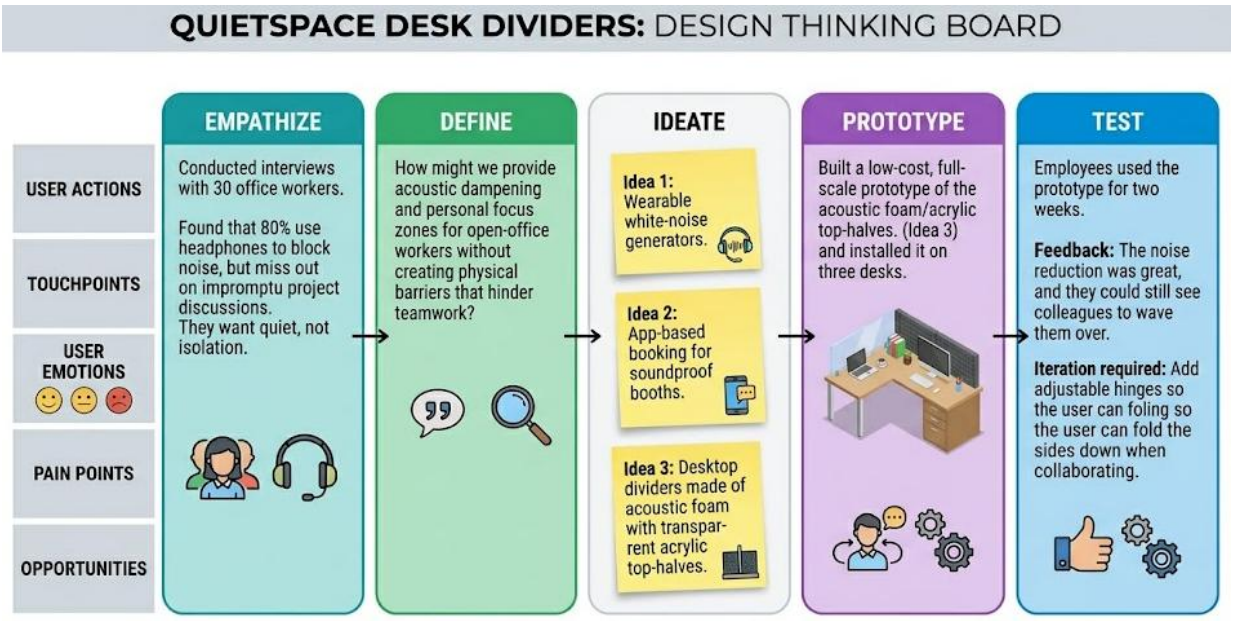
2. **Agile Board :**

Backlog	To do	In progress	Review/Testing	Done
Add social media sharing features.	Integrate Google Maps API for travel distance tracking.	Develop barcode scanner to estimate carbon cost of groceries.	QA test the login module for security flaws.	Finalize basic UI wireframes and brand colors.
Create "Leaderboard" for friends.	Design User Profile dashboard UI	Write algorithm to convert driving miles to CO2 emissions.	Peer code review of the database schema.	Setup cloud hosting and GitHub repository.
Implement push notifications for daily logging.				Implement secure User Authentication (OAuth).

Task 2: Design Thinking Board

1. **Fix a Product and Identify Ambiguous Problem:**

- **Product: "QuietSpace"** – Modular desk dividers for open-plan corporate offices.
- **Ambiguous Problem:** Employees report high stress and distraction in open-plan offices due to noise. However, when placed in traditional, fully enclosed cubicles, they report feeling isolated and disconnected from team collaboration. The problem is finding the exact physical balance between acoustic privacy and visual/social accessibility.



Task 3: Inference Mapping

1. Problem and Reference Article:

- **Problem:** High abandonment rates in online checkout processes for software subscriptions.
- **Reference Theme:** *The Impact of UX Simplification and Single-Page Checkouts on E-Commerce Conversions*.

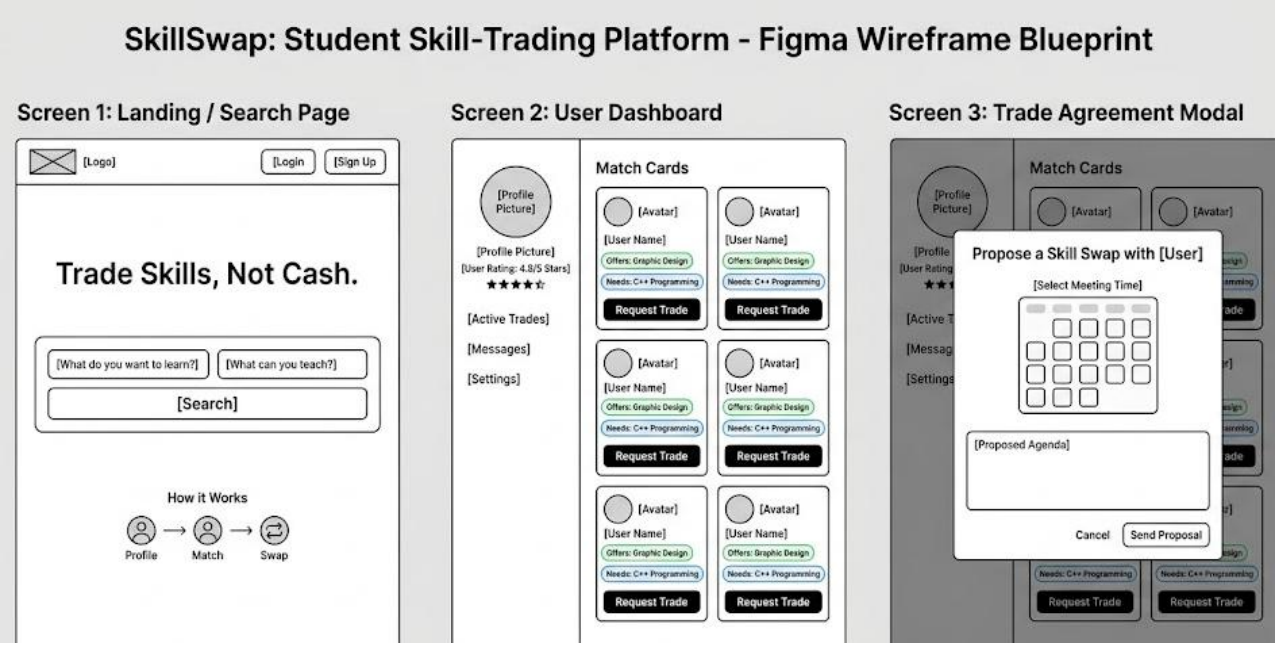
2. Inference Report:

- **Context:** Analytics show that while many users select a software subscription tier, over 65% abandon the purchase when faced with a multi-page checkout requiring account creation, billing details, and address verification on separate screens.
- **The Solution:** The article details a transition to a "Single-Page Checkout" accompanied by a persistent summary of the cart. It also introduces Guest Checkout or one-click OAuth (Google/Apple) logins.
- **Inference & Outcomes:** * Every additional click or page load in a checkout flow increases cognitive load and gives the user time to second-guess the purchase.
    - By collapsing the process into a single, vertically scrolling page with auto-fill capabilities, friction is drastically reduced.
    - **Result:** The inference is that simplifying user interaction directly correlates with revenue growth, often recovering up to 30% of previously abandoned carts.

Task 4: Prototyping Model

1. Select a Product: "SkillSwap" . A web application platform where university students can trade professional skills (e.g., "I will tutor you in Python if you tutor me in Spanish") without money exchanging hands.

2. Design a Solution (Figma Wireframe Blueprint):



Task 5: Case Study Analysis

1. Research Example: IBM (International Business Machines Corporation). IBM successfully combined Agile development with their own framework called "Enterprise Design Thinking."

2. Inference Report:

- **Methodologies Used:**
    - **Enterprise Design Thinking:** IBM trained hundreds of thousands of employees to focus on user outcomes (called "Hills") rather than just technical features. They utilized continuous "Sponsor User" feedback loops.
    - **Agile/Scrum**: Development teams worked in rapid sprints to build out the features dictated by the Design Thinking research, ensuring quick iteration.
- **Challenges Faced:** Cultural Shift: Moving a massive, historically engineering-driven company to a user-centric mindset. Engineers were used to building complex systems first and asking if users wanted them later.
    - **Scaling:** Applying nimble Design Thinking and Agile practices across a highly distributed, global workforce of over 300,000 employees.
- **Outcomes Achieved:** Reduced Development Time: By strictly defining what the user actually needed through Design Thinking *before* writing code, IBM avoided building useless features, cutting time-to-market significantly.
    - **ROI Increase:** According to a Forrester study on IBM's transformation, projects utilizing this combined methodology saw a 300% return on investment and reduced product design time by 75%.

Conclusion: Combining Agile (how to build fast) with Design Thinking (knowing what to build) prevents teams from rapidly developing a product that no one actually wants.